Fiscal Year:	FY 2009	**Task Last Updated:**	FY 04/06/2009
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Project Title:	Mechanisms of HZE Particle-Induced Genetic Instability/Oncogenic Transformation and Their Prevention		
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Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Radiation biology		
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Space Biology Element:	None		
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Comments:	PI moved to Duke University in December 2011. Formerly at University of Colorado Denver (Ed., 2/8/2012)		
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Space Biology 02-OBPR-02
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No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
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Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Received NCE until 11/30/2008 per PI (6/08)		
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	The overall goal of this proposal is to carry out mechanistic studies of genetic instability/neoplastic transformation induced by HZE particles and to develop potential approaches to reduce/prevent it. We will initially focus our efforts on characerizing the phenomenon of HZE particle induced persistent genetic instability in mammalian cells. Next, we will evaluate the potential relationship between genetic instability and oncogenic transformation. Lastly we will explore the potential roles of oxidative stress in HZE particle induced genetic instability/oncogenic transformation and to evaluate if anti-oxidative agents can be effective countermeasures against genetic instability/oncogenic transformation. We have three objectives: 1) To characterize genetic instability induced by HZE particles in mammalian cells. This is to test the hypothesis that HZE particle-induced genetic instability is qualitatively similar to those induced by low-LET radiation. 2) To study the potential role of HZE particle-induced persistent genetic instability in oncogenic transformation of		

Task Description:	exposed cells. This is to evaluate the hypothesis that HZE particle-induced genetic instability will significantly enhance oncogenic transformation. 3) To study the role of oxidative stress in HZE particle induced genetic instability/oncogenic transformation. Both genetic and chemical approaches will be used to study the role of ROS(reactive oxygen species). This is to test the hypothesis that ROS plays major roles in HZE particle-induced genetic instability/oncogenic transformation, and the instability/transformation can be reduced/prevented by the suppression of ROS. The development of malignant diseases is one of the consequences of exposure to cosmic galactic rays of most concern. The experiments to be conducted in this project will evaluate the risks in this respect. Specifically, we will explore the relationhip among HZE particle induced oxidative stress, genetic instability, and oncogenic transformation. The results from this study will faciliate a better understanding of the oncogenic risks associated with exposure to cosmic galactic rays and provide directions for the development of countermeaures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our study have provided a comprehensive estimate of genetic instability induced by HZE particles and the potential involvement of reactive oxidative stress in this important process. It can potentially lead to countermeasures that can reduce/prevent the genotoxic and carcinogenic effects of HZE particle exposures suffered by astronauts. These results will facilitate a better understanding on the relationship between genetic instability and oncogenic transformation , which will benefit the general population on earth.
Task Progress:	I. BRIEF SUMMARY OF THE OBJECTIVES OF THE PROJECT. The overall goal of this proposal is to carry out mechanistic studies of genetic instability/neoplastic transformation induced by HZE (highly charged heavy ions) particles and to develop potential approaches to reduce/prevent it. Specifically, we have the following objectives: • To characterize genetic instability induced by HZE particles. We will study persistent genetic instability in cells that have been exposed to HZE particles. • To study the potential role of HZE particle-induced persistent genetic instability in oncogenic transformation of exposed cells. We will compare the frequency of oncogenic transformation between the clones that exhibit a persistently elevated rate of genetic instability and those that do not. The endpoints will be foci formation in Petri dishes, ability to grow in soft agar, and tumor formation in nude mice. • To study the role of oxidative stress in HZE particle induced genetic instability. We will study the role of the ROS inhibitors in HZE particle induced genetic instability, oncogenic transformation. Specifically, we will study the roles of cellular anti-oxidant genes such as superoxide dismutase (SOD) and explore the effects of chemical anti-ROS agents on HZE particle induced persistent genetic instability. In summary, the proposed study will provide a comprehensive estimate of genetic instability induced by HZE particles and the potential involvement of reactive oxidative stress in this important process. It can also lead to potential countermeasures that can reduce/prevent the genotoxic and carcinogenic effects of HZE particle exposures suffered by astronauts. II. SUMMARY OF ACCOMPLISHMENTS IN THE GRANT PERIOD. 1. Technical accomplishments: in addition to standard assays for evaluating DNA damage, we have established the following assays that can monitor genetic instability more accurately: * A GFP-based deletion reporter that allow us to monitor deletion of the whole reporter gene in mammalian cells. * A GFP-based mismatch repair reporter that can detect either small intragenic mutations or deletions in di-nucleotide repeat in the reporter gene. This assay should be very suitable for the rapid detection of acute or persistent genetic instability induced by HZE particles. Please see section 3E for more details. 2. We have participated in four runs of experiments from 2004-2006 with (0.6 and 1GeV) Fe ions at NSRL facility located in Brookhaven National Laboratory. Most of the planned experiments were conducted successfully. 3. With these runs and additional experiments/analyses that were conducted in the PI's laboratory we have obtained the following key results: A. When measuring acute genetic instability, the micronucleus (MN) assays were used. We have used NIH3T3 cells with over-expression of the SOD1&SOD2 genes to examine the potential anti-mutagenic potential of these two enzymes. Compared to control cells, HZE radiation-induced MN formation in the SOD1 and SOD2- expressing cells are not different statistically. These results are puzzling. However, they suggest that inceasing the levels of SOD enzymes genetically has little effect on radiation-induced genetic instability in these cells. B. We have also used an siRNA-based knockdown approach to examine the roles of Cn/ZnSOD (SOD1) and MnSOD (SOD2) in cellular response to HZE radiation. Our results indicate that down-regulating SOD1 levels has little or no effect on HZE radiation-induced micronucleus formation in murine mammary epithelial cells while down-regulating MnSOD (SOD2) can significantly increase HZE radiation induced micronucleus formation, indicating the differential role of the two SOD enzymes in cellular defense against HZE radiation. C. Our siRNA knockdown results were confirmed also by use of a MnSOD mimetic MnTE-2-Py5+. Our results indicate the addition of the SOD mimetic can reduce Fe ions (1 GeV or 0.6 GeV) induced micronucleus formation in a small but statistically significant fashion. These data are consistent with the idea that acute DNA damage is most caused by direct ionizations of DNA by HZE particles, in contrast to low LET radiation, where most initial damage is caused by secondary free radicals generated by radiation. D. The results on acute DNA damage as measured by micronucleus formation is contrasted with results obtained with measuring long-term DNA damage, which was conducted by use of the GFP-reporter method as mentioned in A. When 4T-1muring mammary cancer cells stably transduced with a single copy of the enhanced green fluorescence protein were irradiated and observed for persistent genetic instability, i.e., GFP gene loss >20 cell divisions later, a significant

effect of MnSOD mimetic MnTE-2-PyP5+ .

From the results obtained, we can see that there was a significant fraction (15.9%) of 4T1 cells that showed chimerical patters at least 23 cell divisions after initial HZE irradiation. This chimeric pattern indicates an ongoing loss of EGFP and provided the strongest proof of persistent genetic instability. In contrast to the acute DNA damage data, we see very significant effects from the MnSOD mimetic MnTE-2PyP5+, which reduced the fraction of chimeric clones from 15.9% to 5.3%. Therefore, free radicals play a significant role in HZE radiation induced persistent genetic instability. This results not only demonstrated the importance of free radicals in mediating longer-term, persistent genetic instability but also suggested promising mitigating roles of MnSOD mimetics for space radiation-induced persistent genetic instability and carcinogenesis.

E. We have also used the BigBlue reporter system that can accurately evaluate small intragenic deletions to complement our EGFP and MN-based approaches which mainly measure large scale events such as gene deletions. This system mainly measures point mutations and small intragenic deletions/insertions. The BigBlue system consists of bacterial phage-based reporter genes that stay silently in rodent (rat or mouse) genome. It is thus susceptible to genetic instability in the host genome. The major advantage is that the after exposure to mutagenic agents, there is no further culturing or expression period needed. Extraction of host DNA and packaging with commercially available packaging extract allow accurate and quantitative measurement of the mutation suffers in the reporter. As delayed type mutations have been associated with oxidative stress related mutations, we thought this system is well suited for our studies despite some concerns about its utility. Our results with the HZE particles proved this to be true. Our results using the rat MEF cell line with the BigBlue reporter gene indicated an elevated frequency of mutation mutations in the surviving clones. Of the 8 survivor colonies analyzed, the mutation frequency ranged from 2.0x10-5 to 2.0x10-4 compared with the control frequency of 8x10-6, indicating significant up-regulation of mutation rate by HZE particles in these cells. These are very exciting data since they indicate that smaller scale mutations were also induced in a persistent manner by HZE radiation.

F. We have developed a novel GFP and firefly luciferase-based mismatch repair reporter system where mutations in microsatellite sequences can be detected by EGFP fluorescence through either flow cytometry, fluorescence microscopy, or bioluminescence . We have used this system to evaluate X-ray induced mutagenesis and preliminary results indicate the effectiveness of the approach. We plan to use this system to evaluate if microsatellite mutations are significantly affected by HZE particles in our future studies with NASA projects.

Bibliography Type:	Description: (Last Updated: 10/30/2019)
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